A possible solution: QKV-Norm

A natural fix could be to apply the normalisation layer *after* the linear operators. In practice this means that we normalise the {query, key, value} vectors, called `QKV-Norm` and defined in section 4.

Paper strategy

Our work is based on three key observations: (i) semantic subspaces are observed in known circuits, (ii) they contribute to the model behaviour, and (iii) `Pre-Norm` requires them to follow a strict latent embedding structure or else interfere through the L_2-norms of the {query, key, value} vectors.

However, it is difficult to demonstrate specific examples of subspace interference. Firstly, a fully-converged model should learn to manage interference for in-distribution examples. Instead, we expect it to concern (i) training stability, (ii) model inductive bias, and (iii) out-of-distribution behaviour. Secondly, circuit explainability is difficult, only being achieved in simple cases. In general we expect circuits to become complicated, contain steps that are harder to interpret than match&pass, and exploit non-interpretable latent subspaces. Difficulty is further increased by polysemanticity [23], the ability for heads and features to change behaviour according to context.

In this work, we take an abstract approach instead. We formally define latent semantic separability, then investigate the theoretical consequences for `Pre-Norm` architectures if this behaviour is important generally. This allows us to make testable predictions about representation structure and model stability without needing to fully reverse-engineer a network or explain subspaces in human terms. We then place some data-driven limits on the effect size. Nonetheless, direct observation remains important, and we hope that future works can confirm or falsify the importance of the proposed representation structure and interference effect.

3 Related Works

Our work is motivated by transformer circuit discovery [8–10, 24–26] and formation [27, 28]. See [11] for a recent review of interpretability for language decoder models, with a list of known logical operations implemented by attention heads. This builds upon works in BERTology [13, 29]. We study normalisation, for which several formulations have been proposed [17, 30–33]. Our `QKV-Norm` variant is similar to `QK-Norm`, which is studied by [18–20] for asymptotic performance and training stability at large learning rates. These are motivated by logit-regularisation, whereas we are motivated by representational inductive bias and stability to latent semantic interference.

We highlight other works that study transformer normalisation through its geometric interpretation as a projection onto a sphere. [34] investigated the role of normalisation in mixing the attention output with the residual stream in `Post-Norm` models, but does not consider `Pre-Norm`. [35] studies the computational abilities of `Pre-LayerNorm` architectures, in particular demonstrating that projection onto a sphere ensures that all keys reside on their own convex hull, preventing them from becoming "unselectable". [36] interprets the latent embeddings of `Pre-Norm` models as a trajectory on a sphere. These works do not consider the interference of semantic subspaces. [37] and the contemporary work [38] study the role of `LayerNorm` in the related phenomenon of embedding rank collapse.

We highlight the contemporary work of [39], who also study multi-step contextual reasoning in transformers using matching operations over independent subspaces, for both `Pre-Norm` and `Post-Norm`. This builds upon [40], who study the learning of abstract symbolic reasoning in transformers, and works that manipulate the flow of information to promote algorithmic reasoning, e.g. [41].

We are not aware of previous works that study the impact of `Pre-Norm`'s spherical geometry on the structure of latent subspaces. However, many works consider linear subspaces, described in the following paragraph. These results are directly applicable to the `No-Norm` and `QKV-Norm` methods in this work, although `QKV-Norm` applies a subsequent spherical projection. [42] design subspace separability into their model by decoupling the normalisation layers for different mechanisms.

Works on vector embeddings [43–45] and the *linear representation hypothesis* [46–48] study the emergence of linear subspaces that encode separable concepts in embedding-unembedding models, using both interpretation and intervention techniques. Many works search for linear subspaces/directions in a transformer representation (e.g. linear probes [49, 50]) or search for faithful causal abstractions (e.g. [51]), with a survey provided in [11] sections 3-4. We also highlight works that study the use

of features in linear superposition [23, 52]. This allows a model to store more features than it has dimensions, at the cost of interference in their linear projections.

The terminology of *semantic subspaces* is used more generally, e.g. [49, 53, 54]. We consider a definition that does not require humans to define the separable concepts, only that abstract latent features remain independent in an attention layer. We also highlight works that study subspaces of static (model input) and contextual (latent or model output) embeddings in transformers, e.g. [54–62] (review in [13]). These are relevant because they also decompose embeddings into a combination of abstract subspaces, capturing different semantic and syntactic structures in a natural language setting. These may be used as semantic subspaces in our work. We highlight [57] which studies interference between positional and contextual components using a decomposition similar to ours, and also experiments using a next-token addition task.

4 Formulation

Consider the `No-Norm` case. Let $\mathcal{X}$ be an unordered set of message receiving tokens, and $\mathcal{Y}$ the message senders. Let $x \in \mathbb{R}^{N_x}$ be the N_x-dimensional representation of an element in $\mathcal{X}$, and $y_t \in \mathbb{R}^{N_y}$ be the tth element in $\mathcal{Y}$, with $1 \leq t \leq T$. For self-attention we have $\mathcal{X} = \mathcal{Y}$. Let $W_Q \in \mathbb{R}^{N_{qkv} \times N_x}$ and $W_K \in \mathbb{R}^{N_{qkv} \times N_y}$ be the query and key weight matrices, with associated vectors $q = W_Q x \in \mathbb{R}^{N_{qkv}}$ and $k_t = W_K y_t \in \mathbb{R}^{N_{qkv}}$ on an N_{qkv}-dimensional latent space. We do not include biases in $\{q,\ k_t\}$ because they contribute terms that are nullified by the `softmax`, or are reproduced by constant directions in x (Theorem 12). We define dot-product attention *scores* as:

$$w_t \ = \ q^T k_t \ = \ x^T W_Q^T W_K y_t \ = \ x^T W_{QK} y_t \tag{3}$$

where $W_{QK} \triangleq W_Q^T W_K \in \mathbb{R}^{N_x \times N_y}$ is a matrix with $Rank(W_{QK}) \leq \min(N_x, N_y, N_{qkv})$. This is the maximum span of the attended subspace in $\{x, y_t\}$. The attention *weights* are

$$a_t \ = \ \texttt{softmax}\,(w_t) \ = \ \frac{e^{w_t}}{\sum_{t'} e^{w_{t'}}} \quad . \tag{4}$$

Let $v_t = W_V y_t \in \mathbb{R}^{N_x}$ be the value vectors with $W_V \in \mathbb{R}^{N_{qkv} \times N_y}$. We do not include biases in v_t because they carry no dependence on the attended token. Each token emits the message $m_t = W_O v_t \equiv W_O W_V y_t \triangleq W_{OV} y_t$ where $W_O = \mathbb{R}^{N_x \times N_{qkv}}$ is the output-matrix. Each attention-head updates x by adding the attention-weighted convex combination of messages, $x \to x + \Delta x$ with $\Delta x = \sum_t a_t m_t$. We usually run H attention-heads in parallel, giving the total update:

$$x \ \to \ x \ + \ \sum_{h=1}^{H} \sum_{t=1}^{T} a_t^{(h)} m_t^{(h)} \qquad \text{Multi-head attention} \tag{5}$$

with unique weights $\{W_Q^{(h)}, W_K^{(h)}, W_V^{(h)}, W_O^{(h)}\}$ for each head index h.

We now introduce normalisation layers. Let $z \in \mathbb{R}^{N_z}$ be any N_z-dimensional vector, then $\texttt{N}(z; \alpha_z) : \mathbb{R}^{N_z} \to \mathbb{R}^{N_z}$ is a normalisation function with parameters α_z. We consider two such functions:

$$\texttt{RMSNorm}\,(z;\ \alpha_z) = \frac{\sqrt{N_z}}{|z|} \text{diag}\,(\alpha_z)\, z \qquad \texttt{LayerNorm}\,(z;\ \alpha_z) = \frac{\sqrt{N_z}}{|z_\perp|} \text{diag}\,(\alpha_z)\, z_\perp \tag{6}$$

[21, 22] where $P_\perp \triangleq \text{diag}\left(1^{N_z}\right) - 1^{N_z} {1^{N_z}}^T$ is a linear operator that subtracts the mean of z from every component, 1^{N_z} is vector of ones, and $z_\perp \triangleq P_\perp z$ is the component of z perpendicular to 1^{N_z}.

The `Pre-Norm` strategy means applying normalisation to the inputs $\{x, y_t\}$. The `QKV-Norm` strategy means applying normalisation to the vectors $\{q, k_t, v_t\}$. We then have three cases:

		w_t	v_t	Norm params
	`No-Norm`	$x^T\ W_{QK}\ y_t$	$W_V\ y_t$	-
(baseline)	`Pre-Norm`	$\texttt{N}\,(x; \alpha_x)^T\ W_{QK}\ \texttt{N}\left(y_t; \alpha_y^K\right)$	$W_V\ \texttt{N}\left(y_t; \alpha_y^V\right)$	$\{\alpha_x,\ \alpha_y^K,\ \alpha_y^V\}$
(alternate)	`QKV-Norm`	$\texttt{N}\,(W_Q x; \alpha_q)^T\ \texttt{N}\,(W_K y_t; \alpha_k)$	$\texttt{N}\,(W_V y_t; \alpha_v)$	$\{\alpha_q,\ \alpha_k,\ \alpha_v\}$

We note that several of these degrees of freedom are redundant and could be combined, e.g. α_q and α_k. We do not consider these variations (i) because they are not relevant for the results of this paper, and (ii) to standardise the number of training parameters.

5 Theory: representation structure required for independent subspaces

Let $\mathbb{S}^N \equiv \mathbb{R}^N$ be an N-dimensional latent representation of $\mathcal{X}$ or $\mathcal{Y}$.

[Definition] Semantic subspace: any independent N_α-dimensional subspace $\mathbb{S}_\alpha^{N_\alpha} \subset \mathbb{S}^N$ for which every element may be uniquely identified by some parameters θ_α, such that it is possible for the attention scores w_t to be fully specified by θ_α. **Semantic separability:** ability for parallel heads to be fully specified by different semantic subspaces.

Let $\{\alpha\}$ be the set of indivisible semantic subspaces. This can be seen as a *co-ordinate system* for the attendable embedding space. Semantic separability requires that each co-ordinate α be independently measurable by an attention head. Let $\mathbb{S}^N$ contain N_s indivisible semantic subspaces $1 \leq \alpha \leq N_s$. Then $\mathbb{S}^N = \prod_\alpha \mathbb{S}_\alpha^{N_\alpha} \bigoplus \mathbb{S}_{\text{null}}$ such that $\sum_\alpha N_\alpha \leq N$ satisfies semantic separability, where $\prod_\alpha, \bigoplus$ are Cartesian products and $\mathbb{S}_{\text{null}}$ is a separable space of non-attended information.

The following theorems derive the representation structure required for semantic separability:

Semantically separable representation structures *[Proofs in appendix F]*

Theorem 1. *`No-Norm`: If two heads with finite non-zero temperature attend to different semantic subspaces, the subspaces must be linearly independent $\mathbb{S}_\alpha^{N_\alpha} \equiv \mathbb{R}^{N_\alpha}$. Corollary: W_{QK} is a low-rank matrix with (left and right) null-spaces that span all non-attended information.*

Theorem 2. *`Pre-Norm`: Semantic subspaces must be represented as orthogonal spheres $\mathbb{S}^{N_\alpha} \equiv \mathcal{S}^{N_\alpha - 1}$ defined using the L_2-norm. Corollary: if either orthogonality or constant-norm are violated, semantic subspaces interfere through a multiplicative factor on w_t.*

Theorem 3. *`QKV-Norm`: Semantic subspaces must be linearly independent.*

We note that every linear subspace $\mathbb{R}^{N_\alpha}$ has N_α continuous degrees of freedom, whilst $\mathcal{S}^{N_\alpha - 1}$ has only $N_\alpha - 1$, the other being removed by the fixed-norm constraint. The subspace $\mathcal{S}^0$ is allowed and may be seen as a binary variable with values $\pm const_\alpha$, and the total representation can store N_s such variables. For `QKV-Norm`, we note that the residual subspace $\mathbb{R}^{N_\alpha}$ only contributes $N_\alpha - 1$ continuous degrees of freedom to the attention calculation, because we apply the projection $\mathbb{R}^{N_\alpha} \to \mathcal{S}^{N_\alpha - 1}$ after extracting the subspace. Table 1 provides a summary.

Structure of messages

We note the special case of compositional annotation, in which a layer creates a semantic subspace that is extracted by a later layer. This is used by circuits including the *induction circuit* [9] described in section 2. By normalising the inputs, `Pre-Norm` induces a *spheroid* message structure close to the *sphere* required for separability in later layers. This may facilitate compositional annotation, aiding in circuit-formation. Message structures are summarised in Table 2.

Strategy	$\mathbb{S}^N$	Representation structure	Attendable d.o.f.
`No-Norm`	$\prod_\alpha \mathbb{R}^{N_\alpha}$	Linearly independent subspaces	N
`Pre-LayerNorm`	$\prod_\alpha \mathcal{S}^{N_\alpha - 1}$	Orthogonal spheres $\perp 1^N$	$N - N_s - 1$
`Pre-RMSNorm`	$\prod_\alpha \mathcal{S}^{N_\alpha - 1}$	Orthogonal spheres	$N - N_s$
`QKV-Norm`	$\prod_\alpha \mathbb{R}^{N_\alpha}$	Linearly independent subspaces	$N - N_s$

Table 1: Representation structure required for semantic separability; *d.o.f.* means *degrees of freedom*.

Strategy	m_t	Structure of m_t	Compositional annotation if
`No-Norm`	$W_{OV} y_t$	Linear	m_t on independent subspace
`Pre-Norm`	$W_{OV} \mathtt{N}(y_t; \alpha_v)$	Spheroid	m_t on orthogonal sphere
`QKV-Norm`	$W_O \mathtt{N}(W_V y_t; \alpha_v)$	Spheroid	m_t on independent subspace

Table 2: Summary of message structures induced by different placements of normalisation layer.